

Pathomporn Niamraksa

☎ 09-8494-9843 | ✉ yuki.snowy@yahoo.com

SKILLS

Programming Languages : C/C++ (OpenGL, Vulkan), Python (NumPy, Matplotlib, OpenGL, multiprocessing)

Development & Build Tools : Visual Studio Code, Git, CMake, GCC

Operating System : Windows, Linux

Other Tools : LaTeX, Google Docs, Google Sheets, Google Slides, GIMP (Basic), Inkscape (Basic), TeamViewer, Oracle VirtualBox (Basic)

EDUCATION

Ramkhamhaeng University

January 2026

Bachelor of Science in Physics : Cumulative GPA 2.08

- **Physics** : Classical Mechanics, Quantum Mechanics, Computational Physics
- **Computer Science** : Object-Oriented Programming, Data Structures, Algorithms, Numerical Methods, Computer Graphics

PROJECTS

Senior Project : Simulation of 2D Brownian Motion and the Fluctuation Theorem in Optical Tweezers

<https://github.com/YukiSnowy/BM-and-FT-in-OT>

- Developed a numerical simulation to model stochastic particle dynamics in an optical trap, solving the Langevin equation using Python.
- Demonstrated the fluctuation theorem by statistical analysis of entropy production in non-equilibrium Brownian dynamics.
- Implemented real-time 2D visualization of stochastic Brownian particle trajectories in an optical trap with Periodic boundary conditions.

RU-Thesis : LaTeX Thesis Template for Ramkamhaeng University

<https://github.com/YukiSnowy/RU-Thesis>

- Developed a custom LaTeX class for university thesis formatting.
- Designed to be easy to use and user-friendly.
- Supports multiple languages, including Thai and English.